

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T03 · Prime Factors, HCF and LCM

With Answers

Questions with full worked solutions

■ 28 pages

FAMILY MAP

Prime Factors HCF LCM

F01 Choose the common-prime envelope

9 Q | 19 marks

F02 Build or manipulate prime-power form

4 Q | 11 marks

Question 02

4MA1-1H Q3

ORIGINAL QUESTION

2 marks

- 3 Find the highest common factor (HCF) of 72 and 108
Show your working clearly.

Worked Solution 02

Highest common factor

MODULAR UNIT 2

2 marks

Prime factors

$$72 = 2^3 \cdot 3^2, \quad 108 = 2^2 \cdot 3^3.$$

Common powers

$$2^2 \cdot 3^2 = 36.$$

Final answer

$$36$$

Question 03

4MA1-1H Q7 demand

ORIGINAL QUESTION UNIT 2

2 marks

$$7 \quad A = 5^3 \times 7^3 \times 11^6 \quad \text{and} \quad B = 5^6 \times 7^2 \times 11^4$$

Find the highest common factor (HCF) of A and B

Give your answer as a product of powers of its prime factors.

.....

Worked solution 03

Highest common factor

MODULAR UNIT 2

2 marks

Demand – Highest common factor

Take minimum exponents

$$\text{HCF} = 5^3 \times 7^2 \times 11^4.$$

Final answer

$$5^3 \times 7^2 \times 11^4$$

Question 07

4MA1-1H Q8

ORIGINAL QUESTION

2 marks

- 8 (a) Write 300 as a product of its prime factors.
Show your working clearly.

.....

(2)

Worked Solution 07

Prime factorisation

MODULAR UNIT 2

2 marks

Divide into primes

$$300 = 3 \times 100 = 3 \times 2^2 \times 5^2.$$

Final answer

$$2^2 \times 3 \times 5^2$$

Question 08

4MA1-1H Q8

ORIGINAL QUESTION

2 marks

$$A = 2 \times 2 \times 2 \times 3 \times 3 \times 5$$

$$B = 2 \times 2 \times 3 \times 3 \times 3 \times 5$$

- (b) Find the lowest common multiple (LCM) of $5A$ and $7B$
Show your working clearly.

.....

(2)

Worked Solution 08

LCM from prime factors

MODULAR UNIT 2

2 marks

Include the extra factors

$$5A = 2^3 \times 3^2 \times 5^2, \quad 7B = 2^2 \times 3^3 \times 5 \times 7.$$

Take each maximum power

$$\text{LCM} = 2^3 \times 3^3 \times 5^2 \times 7 = 37800.$$

Final answer

37800

Original question 05

4MA1-1H Q6 M01

MODULAR UNIT 2

2 marks

6 $A = 2^3 \times 5^4 \times 7 \times 11$

$$B = 2^2 \times 5^2 \times 7^2$$

$$C = 2^2 \times 5^3 \times 7^4$$

Find the highest common factor (HCF) of A , B and C
Write your answer as a product of prime factors.

.....

Worked solution 05

Find the HCF from prime factors

MODULAR UNIT 2

2 marks

M01 – Find the HCF from prime factors**Take the common primes with minimum powers**The common part is $2^2 \times 5^2 \times 7$.**Write the product**

$$\text{HCF} = 2^2 \times 5^2 \times 7 = 700.$$

Final answer

$$2^2 \times 5^2 \times 7$$

Original question 03

4MA1-1H Q3 Q3(a)

ORIGINAL QUESTION**2 marks**

- 3 (a) Find the highest common factor (HCF) of 56 and 84
Show your working clearly.

(2)

Worked solution 03

Chapter: Prime Factors, HCF & LCM

MODULAR UNIT 2

2 marks

Q3(a) – Chapter: Prime Factors, HCF & LCM

Family: Choose the common-prime envelope • Variant: HCF of ordinary integers

Method $56 = 2^3 \times 7$ and $84 = 2^2 \times 3 \times 7$.*Why: Prime factorisation displays the factors common to both numbers.***Method**The common prime factors with the smaller powers give $\text{HCF} = 2^2 \times 7 = 28$.*Why: The HCF uses only primes shared by both numbers at their minimum exponents.***Final answer****HCF: 28**

Original question 04

4MA1-1H Q3 Q3(b)

ORIGINAL QUESTION**2 marks**

- (b) Find the lowest common multiple (LCM) of 60 and 72
Show your working clearly.

(2)

Worked solution 04

Chapter: Prime Factors, HCF & LCM

MODULAR UNIT 2

2 marks

Q3(b) – Chapter: Prime Factors, HCF & LCM

Family: Choose the common-prime envelope • Variant: LCM of ordinary integers

Method

$$60 = 2^2 \times 3 \times 5 \text{ and } 72 = 2^3 \times 3^2.$$

*Why: Prime factorisation displays every factor needed by either number.***Method**

$$\text{LCM} = 2^3 \times 3^2 \times 5 = 360.$$

*Why: The LCM uses every prime at its greatest exponent.***Final answer****LCM: 360**

Original question 20

4MA1-2H Q7

ORIGINAL QUESTION**3 marks**

- 7 Find the lowest common multiple (LCM) of 28, 42 and 63
Show your working clearly.

Worked solution 20

Chapter: Prime Factors, HCF & LCM

MODULAR UNIT 2

3 marks

Q7 – Chapter: Prime Factors, HCF & LCM

Family: Choose the common-prime envelope • Variant: LCM of ordinary integers

Method $28=2^2 \times 7$, $42=2 \times 3 \times 7$ and $63=3^2 \times 7$.*Why: Prime-factorise each number.***Method**The LCM uses the greatest power of every prime: $2^2 \times 3^2 \times 7$.*Why: Take each prime at its maximum exponent.***Method**

LCM=252.

*Why: Evaluate the product.***Final answer**

LCM: 252

2 Find the lowest common multiple (LCM) of 28 and 105

Worked solution

January 2020 | Question 2 | 2 marks

Family: Choose the common-prime envelope • Variant: LCM of ordinary integers

Method / working

1. $28=2^2 \times 7$ and $105=3 \times 5 \times 7$.

2. $\text{LCM}=2^2 \times 3 \times 5 \times 7=420$.

Final answer: 420

9

$$A = 2^3 \times 3^2 \times 5^2 \times 11$$

$$B = 2^4 \times 3 \times 5^4 \times 13$$

Find the lowest common multiple (LCM) of A and B .
Give your answer as a product of powers of prime numbers.

Worked solution

January 2021 | Question 9 | 2 marks

Family: Choose the common-prime envelope • Variant: HCF or LCM from prime-power data

Method / working

1. Take the greatest prime exponent from A and B for each prime.
2. The LCM is $2^4 \times 3^2 \times 5^3 \times 11 \times 13$.

Final answer: $2^4 \times 3^2 \times 5^3 \times 11 \times 13$

- 1 Write 600 as a product of powers of its prime factors.
Show your working clearly.

Worked solution

June 2021 | Question 1 | 3 marks

Family: Build or manipulate prime-power form • Variant: Prime-factorise an integer

Method / working

1. Divide 600 successively into prime factors: $600=2 \times 300=2 \times 2 \times 150=2 \times 2 \times 2 \times 75$.

2. Continue $75=3 \times 25=3 \times 5 \times 5$.

3. Collect powers: $600=2^3 \times 3 \times 5^2$.

Final answer: $2^3 \times 3 \times 5^2$

- 7 Write 3.6×10^3 as a product of powers of its prime factors.
Show your working clearly.

Worked solution

November 2021 | Question 7 | 3 marks

Family: Build or manipulate prime-power form | Variant: Prime-factorise an integer

METHOD / WORKING

1. $3.6 \times 10^3 = 3600$

2. $3600 = 36 \times 100$

3. $36 = 2^2 \times 3^2, 100 = 2^2 \times 5^2$

4. So

5. $3600 = 2^4 \times 3^2 \times 5^2$

6. Answer: $2^4 \times 3^2 \times 5^2$

FINAL ANSWER

$2^4 \times 3^2 \times 5^2$

10 $A = 2 \times 3^{43}$
 $B = 16 \times 3^{37}$

(a) Find the highest common factor (HCF) of A and B .

.....
(1)

(b) Express the number $A \times B$ as a product of powers of its prime factors.
Give your answer in its simplest form.

.....
(2)

.....
(Total for Question 10 is 3 marks)

Answer reference | May-Nov 2020 | Question 10

Family: Build or manipulate prime-power form • Variant: Combine given prime-power products

10	(a)		2×3^{37}	1	B1
	(b)	$2 \times 3^{43} \times 2^4 \times 3^{37}$ or $2^5 \times 3^p$ ($p \neq 80$) or $2^q \times 3^{80}$ ($q \neq 5$)		2	M1
			$2^5 \times 3^{80}$		A1
					Total 3 marks

Worked solution

May-Nov 2020 | Question 10 | 3 marks

Family: Build or manipulate prime-power form • Variant: Combine given prime-power products

Method

$$A = 2 \times 3^{(43)}$$

$$B = 16 \times 3^{(37)} = 2^4 \times 3^{(37)}$$

For the HCF, take the smaller powers:

$$\text{HCF} = 2 \times 3^{(37)}$$

For $A \times B$,

$$A \times B = (2 \times 3^{(43)})(2^4 \times 3^{(37)})$$

$$A \times B = 2^5 \times 3^{(80)}$$

Answer: (a) $2 \times 3^{(37)}$, (b) $2^5 \times 3^{(80)}$